Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks. We propose the Transformer, a model architecture eschewing recurrence and entirely relying on an attention mechanism to draw global dependencies between input and output.

1. Introduction

Recurrent neural networks, particularly LSTM and gated recurrent neural networks, have been firmly established as state of the art approaches in sequence modeling. The Transformer allows for significantly more parallelization.

3. Model Architecture

Most competitive neural sequence transduction models have an encoder-decoder structure. An attention function can be described as mapping a query and a set of key-value pairs to an output.

Equation 1: Scaled Dot-Product Attention:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}((Q * K^T) / \sqrt{d_k}) * V$$

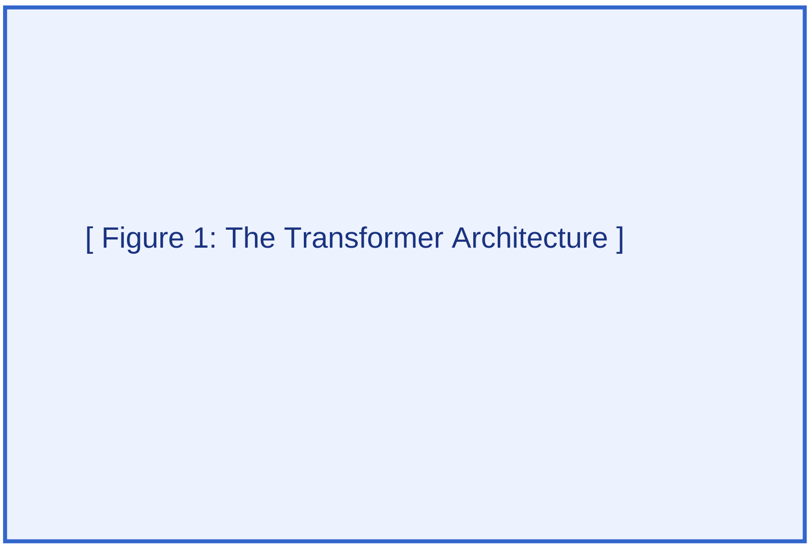

Figure 1: The Transformer - model architecture diagram.